PPO and RLHF: Reference Sheet

Companion to the *Pipeline PPO* code-along. Every equation here appears in the code.

The loop in five steps

1. **Clone a reference policy.** Imitate the playbook (RLHF: supervised fine-tuning). Freeze a copy as π_ref.
2. **Collect experience.** Run the current policy on many deals for a few weeks. Store state, action, log-probability, value and reward.
3. **Score the experience.** Compute advantages with GAE: which plays did better than the forecast expected?
4. **Update.** Several passes of the clipped policy objective, the value loss, an entropy bonus and the KL penalty.
5. **Repeat.** The new policy collects the next batch.

Symbols

SYMBOL	MEANING	IN THIS PROJECT
s, s_t	State	Deal snapshot: stage, fit, engagement, weeks open, size, discount and sponsor flags, last play
a, a_t	Action	One of six plays
r_t	Reward	Revenue at close minus play and holding costs, in $100K
π_θ(a|s)	Policy (actor)	Probability of each play, from a softmax over logits
π_old	Policy that collected the batch	Frozen during the update
π_ref	Reference policy	The playbook clone (RLHF: the SFT model)
V_θ(s)	Value function (critic)	Opportunity value: expected discounted net revenue from s
γ	Discount factor	0.99 per week
λ	GAE parameter	0.95
ε	Clip range	0.2
β	KL coefficient	0.02
c_ent	Entropy coefficient	0.01

Core equations

Discounted return (what actually happened from state s)

$$G_t = r_t + \gamma\, r_{t+1} + \gamma^2\, r_{t+2} + \dots = \sum_k \gamma^k\, r_{t+k}$$

Value loss (trains the critic; the equation from the lesson)

$$L^{VF} = \tfrac{1}{2}\,(V_\theta(s_t) - R_t)^2 \quad \text{where the target } R_t = A_t + V_{old}(s_t)$$

One-step surprise (TD error)

$$\delta_t = r_t + \gamma\, V(s_{t+1})\,(1 - \text{done}_t) - V(s_t)$$

Generalized advantage estimation

$$A_t = \delta_t + \gamma\,\lambda\,(1 - \text{done}_t)\,A_{t+1}$$

$\lambda = 1$ gives the full Monte Carlo return minus V (unbiased, noisy). $\lambda = 0$ gives δ_t alone (stable, biased).

Probability ratio

$$r_t(\theta) = \pi_\theta(a_t|s_t) \,/\, \pi_{old}(a_t|s_t) = \exp(\log \pi_\theta - \log \pi_{old})$$

Clipped policy objective (maximize)

$$L^{CLIP} = \mathbb{E}\left[\, \min\left(r_t\, A_t,\ \text{clip}(r_t,\, 1-\varepsilon,\, 1+\varepsilon)\, A_t \right) \right]$$

Entropy (exploration bonus, maximize)

$$H(\pi) = -\sum_a \pi(a|s)\, \log \pi(a|s)$$

KL to the reference (penalty, minimize)

$$KL(\pi_\theta \,\|\, \pi_{ref}) = \sum_a \pi_\theta(a|s)\, [\log \pi_\theta(a|s) - \log \pi_{ref}(a|s)]$$

Total loss minimized each minibatch

$$L = -L^{CLIP} + c_{vf} \cdot L^{VF} - c_{ent} \cdot H + \beta \cdot KL$$

Gradients written out in the code (softmax policy, logits z)

QUANTITY	GRADIENT WITH RESPECT TO LOGIT Z_J
log π(a)	1[j = a] − π_j
Clipped objective	g · (1[j = a] − π_j), where g = r·A if the unclipped term is the minimum, else 0

QUANTITY	GRADIENT WITH RESPECT TO LOGIT z_j
Entropy H	$-\pi_j (\log \pi_j + H)$
KL to reference	$\pi_j [(\log \pi_j - \log \pi_{ref,j}) - KL]$
Cross-entropy (cloning)	$\pi_j - \text{target}_j$
Value loss	$V - R$, with respect to the critic's output

Reading the training log

SIGNAL	HEALTHY RANGE	WHAT IT TELLS YOU
Net value per deal	Rises above the playbook, then plateaus	The thing you care about
Value loss	Falls, then noisy-flat	How well the critic forecasts
Explained variance	0.4 to 0.9 here	Share of return variation the critic explains (1 = perfect, 0 = useless)
Clip fraction	1% to 20%	How often the clip limited the update. Very high means the learning rate is too aggressive
Approximate KL (old vs new)	Below about 0.02 per update	Size of each policy step
KL to reference	Rises, then levels off	Distance from the playbook. Unbounded growth means the leash is too loose
Entropy	Declines slowly	Exploration. A collapse to near zero means the policy stopped trying alternatives

Hyperparameters and what moving them does

KNOB	RAISE IT	LOWER IT
γ (discount)	Values later revenue more; longer credit assignment	More short-term; less noise
λ (GAE)	Trusts real rewards more; less bias, more variance	Trusts the critic more; more bias, less variance
ε (clip)	Bigger steps per update; faster but riskier	Smaller, safer steps
β (KL)	Stays closer to the playbook	Freer to deviate (and, in RLHF, to exploit reward-model flaws)
c_{ent}	More exploration	Faster commitment, risk of premature collapse
Epochs per batch	Squeezes more from each batch; risk of overfitting it	More stable, needs more data
Learning rate	Faster learning, more instability	Slower, steadier

RLHF mapping

RLHF	PIPELINE PPO
Prompt + tokens so far	Deal state
Next token	Next play
Completion	One deal from open to close
SFT model / reference	Playbook clone
Reward model	Revenue minus cost (here the true reward; in RLHF a learned and imperfect proxy)
Value head on the LLM	Opportunity value
KL penalty per token	KL penalty per weekly decision
Reward hacking	Exploiting a simulator quirk that wouldn't hold in the real market

Results from the reference run

Net value per deal \$15.65K (playbook) → \$18.22K (PPO), +16.4%. Discounted wins 100% → 9%. Value-head R^2 0.44 vs 0.11 for stage averages. KL experiment (120 iterations): $\beta = 0$ → \$17.39K, KL 0.51; $\beta = 0.02$ → \$16.54K, KL 0.30; $\beta = 0.2$ → \$14.50K, KL 0.05.